Zhuhui Tang

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RESEARCH PROFILE

Interdisciplinary researcher with medical training and experience in clinical informatics, real-world clinical data, medical knowledge representation, and the interpretation of multimodal physiological measures.

My research interests lie in human-centered AI for health, particularly how physiological sensing, clinical knowledge, and self-reported experiences can be integrated to make health information more interpretable, contextualized, and actionable.

EDUCATION

China Academy of Chinese Medical Sciences **Beijing, China | Sep 2022 – Jun 2025**

Master of Medicine

Master's Thesis: *A Study on Knowledge Discovery Methodology Based on Knowledge Graph of Traditional Chinese Medicine Records for Insomnia*

Relevant Coursework: Medical Statistics (96/100) **Average Grade: 86/100**

Changsha Medical University **Changsha, China | Sep 2015 – Jun 2020**

Bachelor of Medicine, Traditional Chinese Medicine

Academic Standing: Top 15% of undergraduate cohort

SELECTED RESEARCH EXPERIENCE

Multimodal Physiological Sensing and Clinical Workflow Prototyping | *Project Page*

Research Assistant | *Institute of Information on Traditional Chinese Medicine, China Academy of Chinese Medical Sciences | Oct 2025–Present*

- Participated in outpatient data collection involving ECG, EEG, electrodermal measurements, pulse, respiration, clinical questionnaires, and patient records.
- Applied a MATLAB-based multimodal signal-processing pipeline, gaining practical experience in preprocessing, signal quality assessment, temporal alignment, and feature extraction across physiological modalities.
- Supported the organization of exported and processed physiological data alongside clinical questionnaire records; currently involved in statistical analysis of the collected data.
- Developed a lightweight web-based tool using AI-assisted coding and speech recognition to transform dictated questionnaire responses into structured fields and export Excel records.

Interactive Insomnia Knowledge System | *Project Page*

Co-Principal Investigator | *China Academy of Chinese Medical Sciences | Grant No. YZX-202330 | Completed 2025*

- Co-led the end-to-end research process, from project conception, proposal writing, and grant application to study design and implementation.
- Led clinical knowledge engineering and graph schema design, selecting 296 eligible records from 976 insomnia-related cases and developing terminology dictionaries covering 330 symptom terms and 75 standardized TCM syndrome terms.

- Directed the design and implementation of an interactive knowledge exploration system, integrating graph-based representation of symptom–syndrome–herbal medicine relationships, interactive visualizations, association analysis, and custom-formula exploration.
- Evaluated the proposed methodology by comparing case-derived associations with established domain literature, documenting observed patterns and assessing the feasibility of the knowledge discovery approach.

Research Outputs: Master's Thesis (2025); *Intelligent Knowledge System for Insomnia Management (V1.0)*, Software Copyright.

Clinical Knowledge Infrastructure for Heart-System Diseases

Core Researcher | *China Academy of Chinese Medical Sciences* | *Grant No. YZX-202313* | *2023–Present*

- Designed hierarchical clinical ontology schemas for TCM heart-system conditions, including insomnia, palpitations, and chest impediment, standardizing disease, symptom, and syndrome concepts across heterogeneous clinical records.
- Led the construction of structured clinical datasets, implementing inclusion/exclusion criteria, terminology normalization, and record-to-schema mapping procedures.
- Coordinated technical implementation and data processing workflows, performing data quality checks, consistency reviews, and iterative corrections to prepare analysis-ready datasets for cross-condition research.

SELECTED PUBLICATIONS

Tang, Z., Wu, Q., Jia, Y., et al. (2024). Research progress on knowledge discovery in traditional Chinese medicine medical records based on artificial intelligence. *Guidelines and Standards of Chinese Medicine*, 2(4), 174–181. *Paper*

Tang, Z., Tong, X., Jia, Y., et al. (2024). Reflection on tacit knowledge discovery methods based on clinical cases of traditional Chinese medicine. *Chinese Journal of Basic Medicine in Traditional Chinese Medicine*, 30(11), 1856–1860. (In Chinese)

Yu, J., Tong, X., Tong, L., **Tang, Z**., et al. (2025). Knowledge graph completion technology and its application in mining traditional Chinese medicine classics. *Chinese Journal of Basic Medicine in Traditional Chinese Medicine*, 31(8), 1339–1346. (In Chinese)

SKILLS & CLINICAL QUALIFICATIONS

Programming & Tools: Python, MATLAB, Git, SQL, Neo4j.

Medical Licensure: Licensed Physician (Traditional Chinese Medicine, China).

Clinical Experience: 3+ years of supervised clinical training and patient-facing experience, including care for patients with insomnia, anxiety, and related health concerns.

Additional Certifications: Psychological Counseling; Health Management.

LEADERSHIP & SERVICE

Academic Representative, Master's Cohort of 2022, China Academy of Chinese Medical Sciences.

Class Representative, Institute of Basic Theory for Chinese Medicine, Cohort of 2022.

Volunteer Service: Completed 100+ hours of medical outreach and event hosting.

Award: Graduate Student Volunteer Dedication Award (2024).